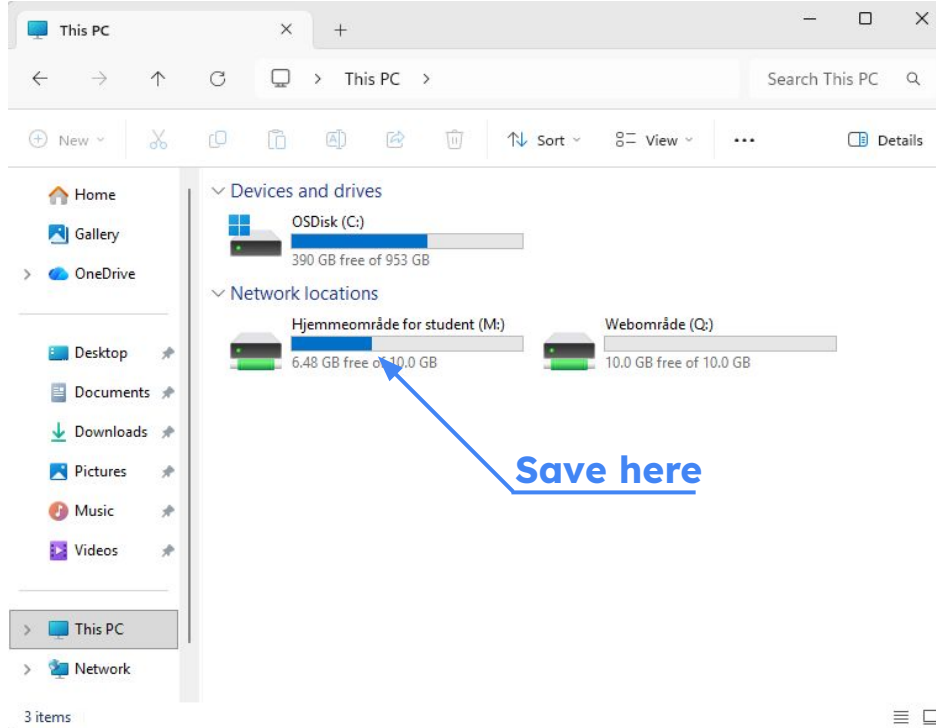


# SAP2000<sup>®</sup>

## Introduction

# Save file



Or:



Google Drive



OneDrive

# Structure/Geometry

---

## **Workflow**

*0. General*

1. Grid
2. Materials
3. Frame Sections
4. Frame objects
5. Connections

# General

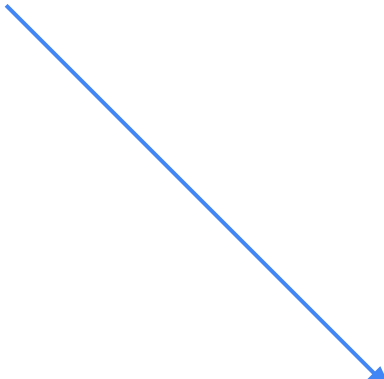
## Cursor

- Pan: middle mouse
- Rotate: shift - middle mouse

## Display options

- Edit => Display options
  - 3D view:  
=> View options=>Extrude

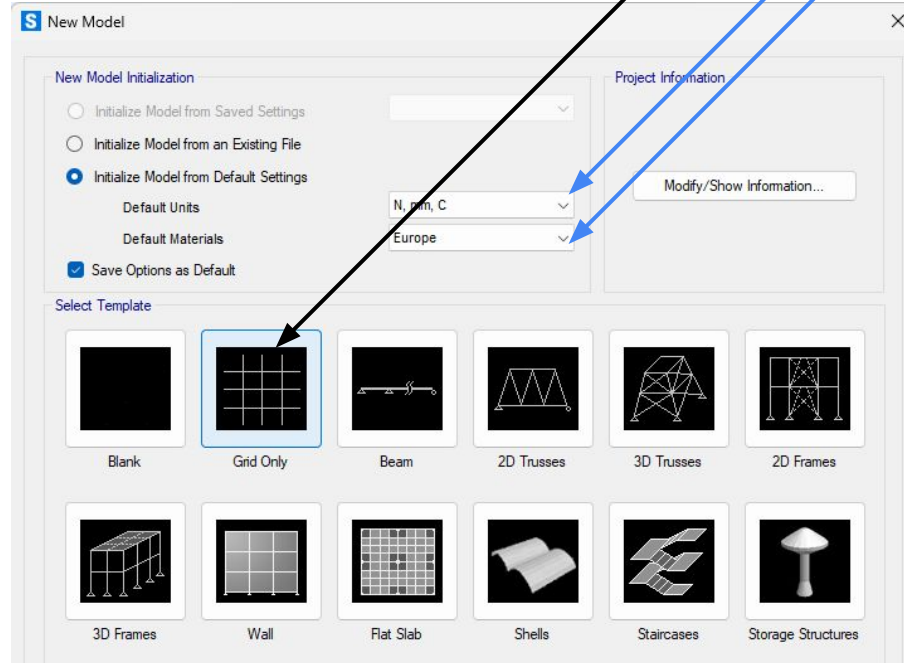
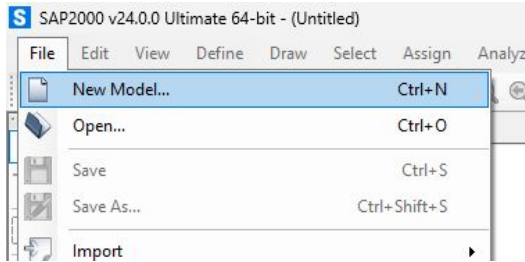
Change units: Bottom  
right



# New model complex geometry

Click Grid Last

N/kN  
Europe



# Grid Setup

## Modify grid later:

1.

**Quick Grid Lines**

Cartesian Cylindrical

Coordinate System Name: GLOBAL

Number of Grid Lines

X direction: 5

Y direction: 10

Z direction: 2

Grid Spacing

X direction: 100

Y direction: 100

Z direction: 100

First Grid Line Location

X direction: 0.

Y direction: 0.

Z direction: 0.

OK Cancel

2.

**Coordinate/Grid Systems**

Systems: GLOBAL

Click to:

Add New System...

Add Copy of System...

Modify/Show System...

Delete System

Convert to General Grid

OK Cancel

3.

**Define Grid System Data**

System Name: GLOBAL

Grid Lines: Quick Start...

Add Delete

Display Grids as: ☒ Ordinates ☐ Spacing

☐ Hide All Grid Lines

☐ Glue to Grid Lines

Bubble Size: 25

Reset to Default Color

Reorder Ordinates

OK Cancel

**X Grid Data**

| Grid ID | Ordinate (mm) | Line Type | Visible | Bubble Loc | Grid Color |
|---------|---------------|-----------|---------|------------|------------|
| A       | 0             | Primary   | Yes     | End        |            |
| B       | 100           | Primary   | Yes     | End        |            |
| C       | 200           | Primary   | Yes     | End        |            |
| D       | 300           | Primary   | Yes     | End        |            |
| E       | 400           | Primary   | Yes     | End        |            |
| F       | 500           | Primary   | Yes     | End        |            |

**Y Grid Data**

| Grid ID | Ordinate (mm) | Line Type | Visible | Bubble Loc | Grid Color |
|---------|---------------|-----------|---------|------------|------------|
| 1       | 0             | Primary   | Yes     | Start      |            |
| 2       | 100           | Primary   | Yes     | Start      |            |
| 3       | 200           | Primary   | Yes     | Start      |            |
| 4       | 300           | Primary   | Yes     | Start      |            |
| 5       | 400           | Primary   | Yes     | Start      |            |
| 6       | 500           | Primary   | Yes     | End        |            |

**Z Grid Data**

| Grid ID | Ordinate (mm) | Line Type | Visible | Bubble Loc | Grid Color |
|---------|---------------|-----------|---------|------------|------------|
| Z1      | 0             | Primary   | Yes     | End        |            |
| Z2      | 100           | Secondary | Yes     | End        |            |
| Z3      | 200           | Primary   | Yes     | End        |            |

Add Line (same spacing)

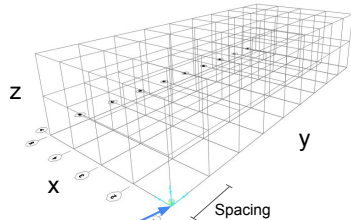
Switch Coordinates/Spacing

Bubble

Primary (w/ Bubble)

Secondary (No Bubble)

Last: OK



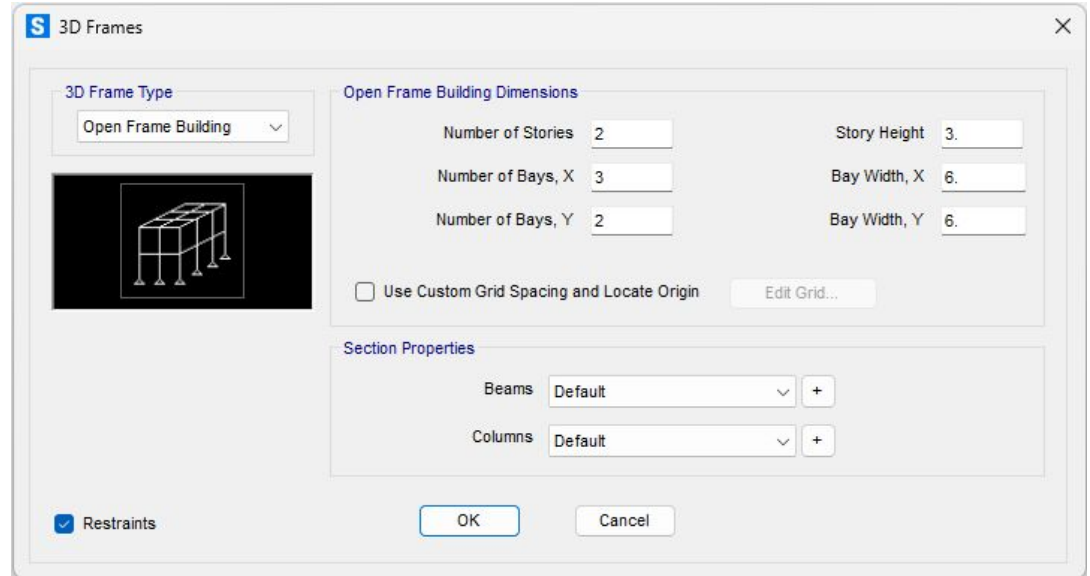
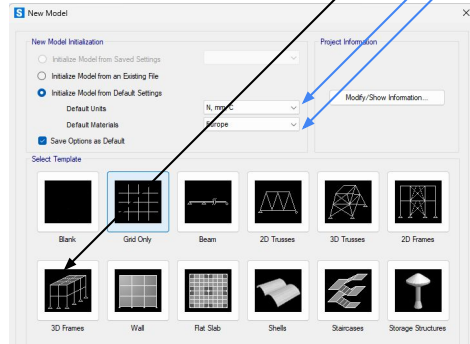
First Grid Line Location (Origin)

# New model simple geometry

Click 3D Last

N/kN  
Europe

Choose stories and bays based on your structure



# Frame objects



## Select object

Select frame or node to edit

## Element drawing options



### Draw frame element

Draw a frame or brace element (Ramme-element/stav).



### Draw 2D surface or square

Use to draw walls/shear walls

## Node snapping options: Choose after element selected



### Snap to **Grid** (intersections)

Snaps to defined grid



### Snap to perpendicular

Snaps to a point perpendicular to the frame element of the first point



### Snap to ends and **midpoints**

Useful for Bracing/fagverk



### Snap to closest (Dont use)

Snaps to closest element/node. Only use for ideas or tests.



### Snap to element intersections

Intersections between frame elements



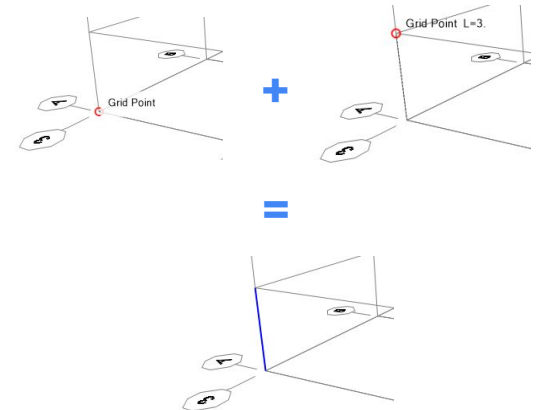
### Snap to increment

Snaps to every 0.1cm etc.

## 1. Choose object

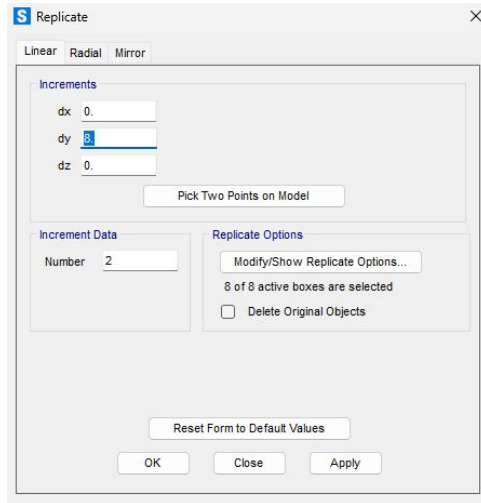
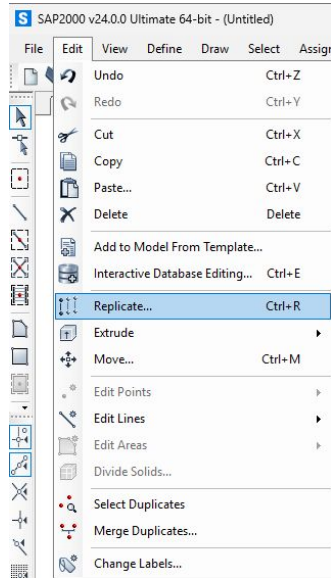
## 2. Choose snapping

## 3. Click on 2 points





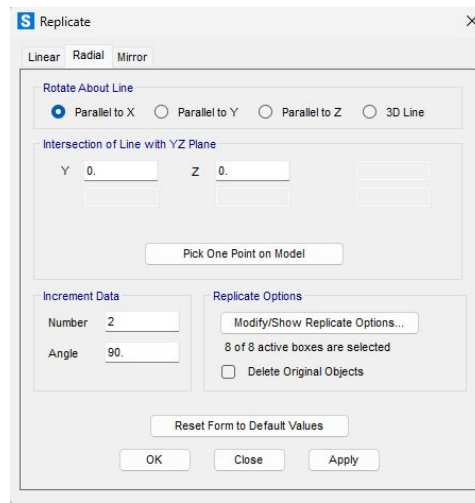
# Mirror/rotate/duplicate



## Duplicate

**Increment:** Choose distance between duplicates

**Increment Data:** Choose number of copies



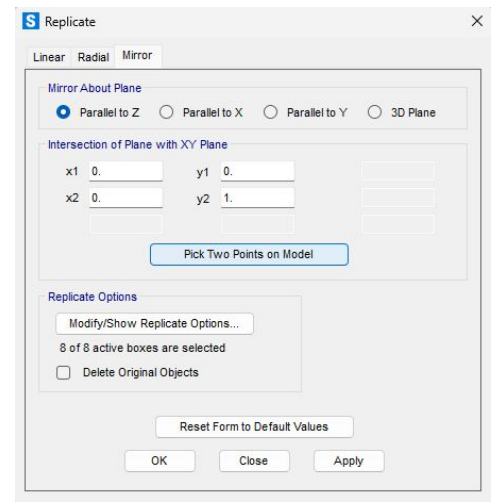
## Rotate

**About axis:** Choose axis to rotate around

**Intersection:** Choose where the axis is

**Increment Data:** Choose number of copies

**Angle:** Choose angle to rotate  
**Options:** Delete original objects if you want to move the selection, instead of making a copy



## Mirror

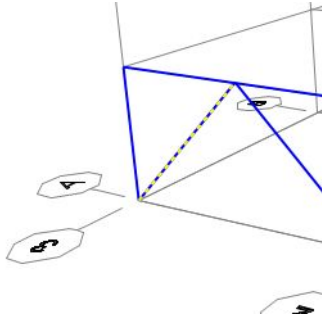
**About axis:** Choose axis to mirror about

**Intersection:** Choose where the plane is

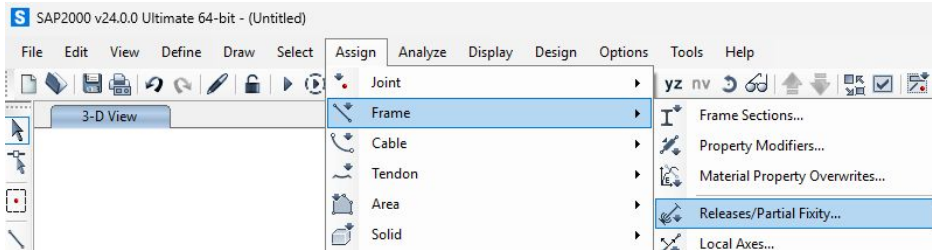
**Delete original objects** if you want to move the selection, instead of making a copy

# Connections

## 1. Choose drawn element (Turns Blue/Yellow)



## 2. Assign => Frame => Releases/Partial Fixity

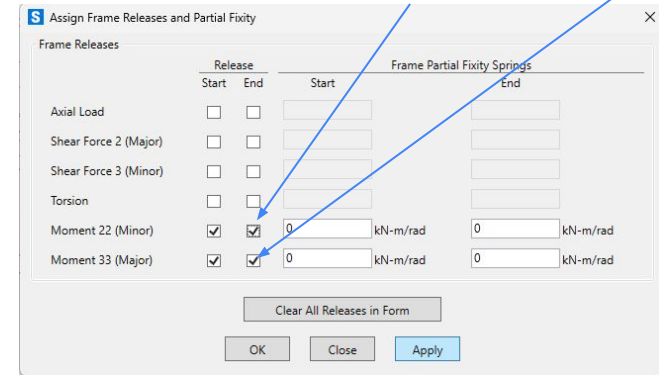
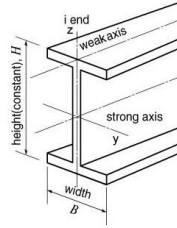


3.

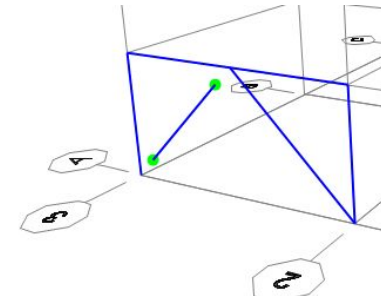
### a) Pinned-Pinned

=> Release Moment,

- Major = y axis, Minor = z axis

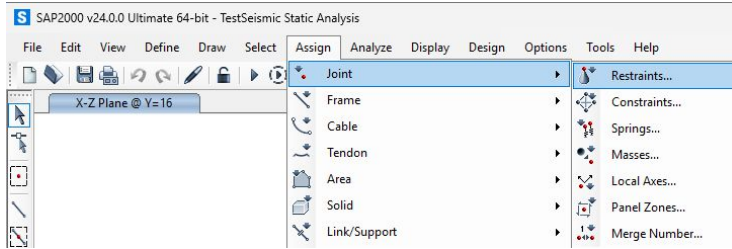


4.

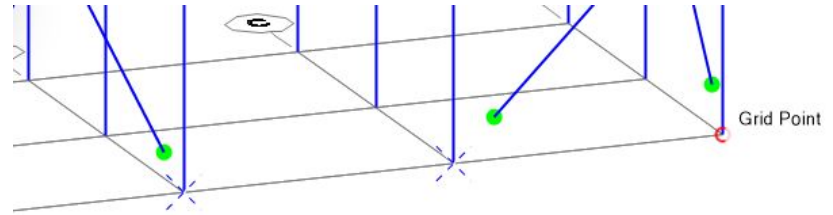


# Boundary Conditions

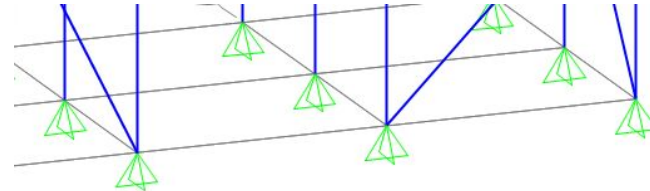
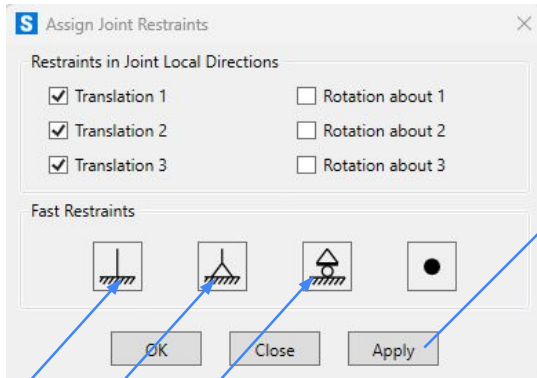
1.



2. Pick all nodes where you want a joint



3. Choose Restraint

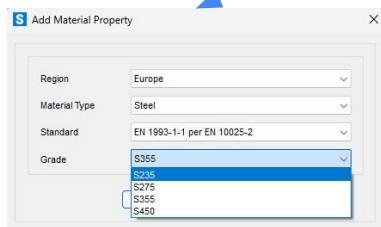
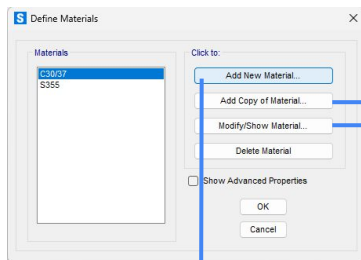


# Materials

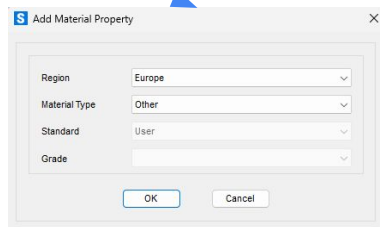
1.



2.

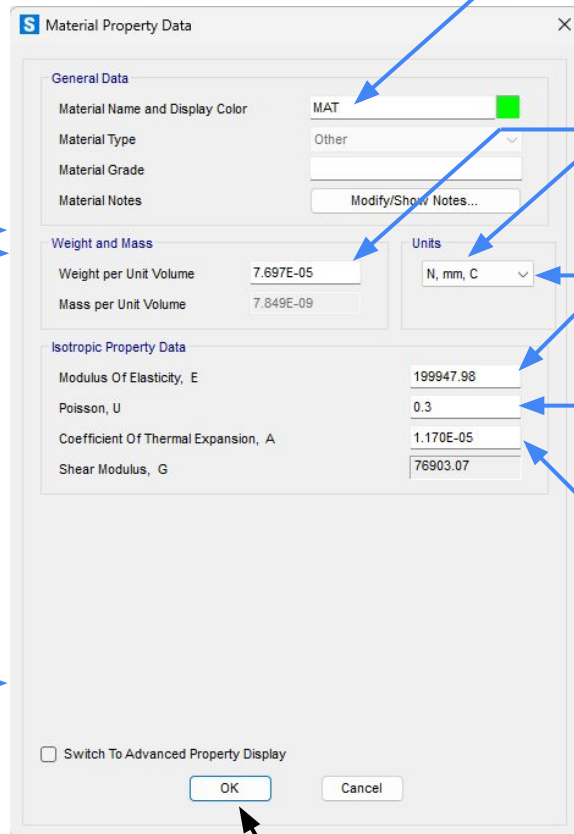


Eurocode Materials



Timber and other materials

3.



Material Name:  
GL24h, C24 etc.

Density  
(NEWTONS)

Watch units

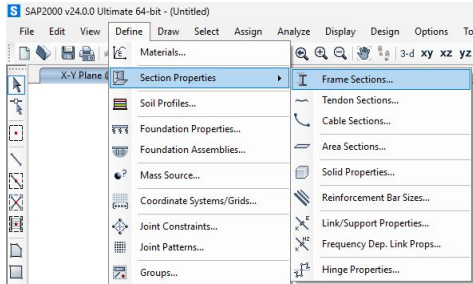
$$\nu = \frac{\epsilon_{\perp}}{\epsilon_{\parallel}}$$

$$\alpha \cdot \Delta T = \epsilon$$

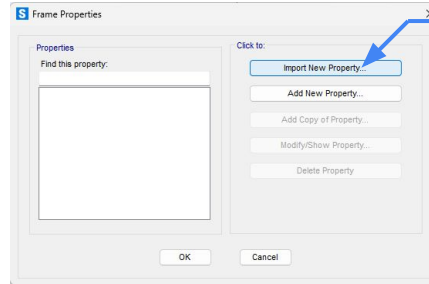
# Frame sections (Cross section/tverrsnitt)

## Complex sections (I, H, C)

1.

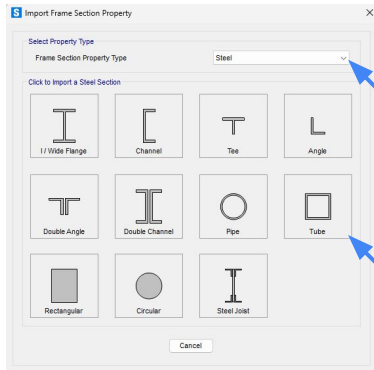


2.



Import  
New

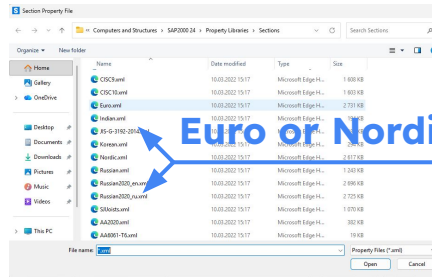
3.



Material

Section

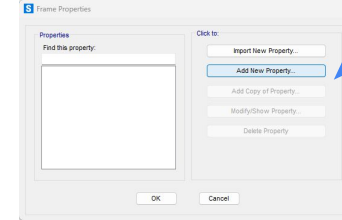
4.



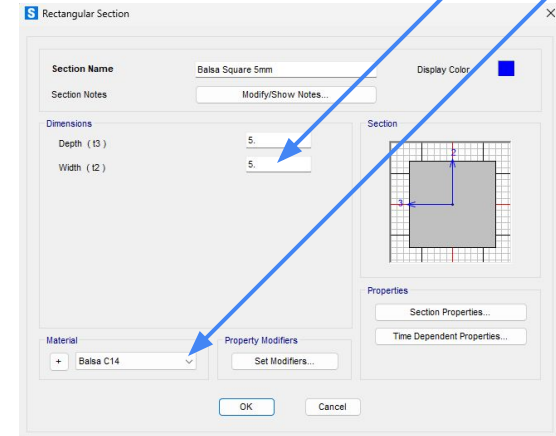
Euro or Nordic

## Simple Sections (or non standard)

Add New



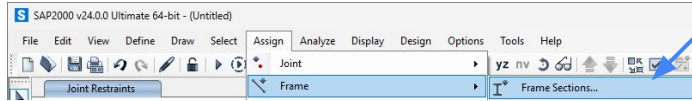
Fill out



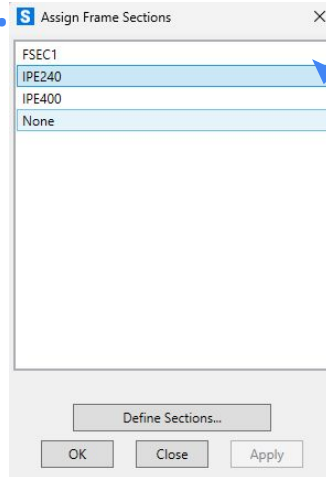
# Assign frame sections

1.

Assign  
→ Frame  
→ Frame Sections

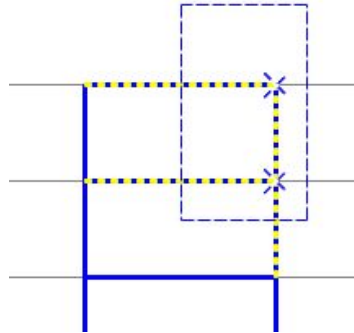


2.

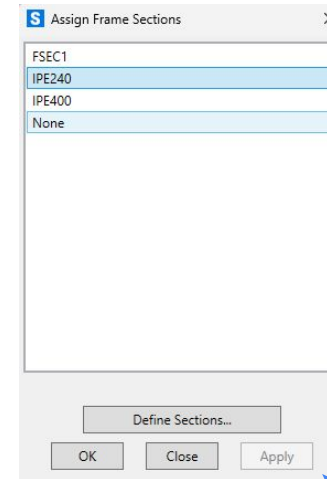


Choose section

3. Select frame elements



4.



Apply

# Static Analysis

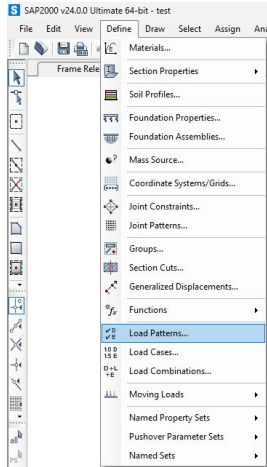
---

## Workflow

1. Load types
2. Optional: Nonlinear analysis
3. Load Combinations
4. Assign Loads
5. Analysis

# Loads

1.



2.

**Name**

**Type**

**If Dead => 1: Auto adds weight**

**Last: Click add**

**Define Load Patterns**

| Load Pattern Name | Type | Self Weight Multiplier | Auto Lateral Load Pattern |
|-------------------|------|------------------------|---------------------------|
| Snow              | Snow | 0                      |                           |
| DEAD              | Dead | 1                      |                           |

Click To:

- Add New Load Pattern
- Add Copy of Load Pattern
- Modify Load Pattern
- Modify Lateral Load Pattern...
- Delete Load Pattern
- Show Load Pattern Notes...

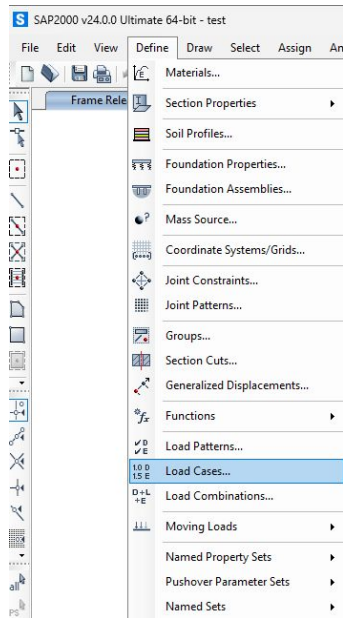
OK Cancel

**OK**

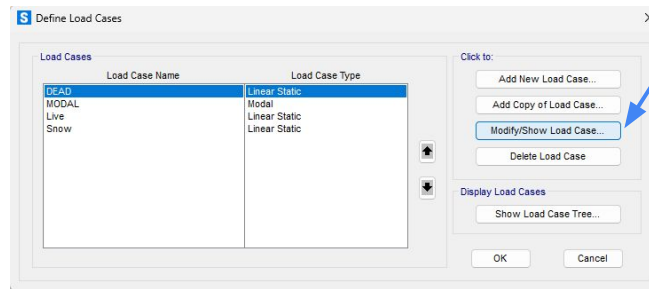


# Non-linear analysis

1.

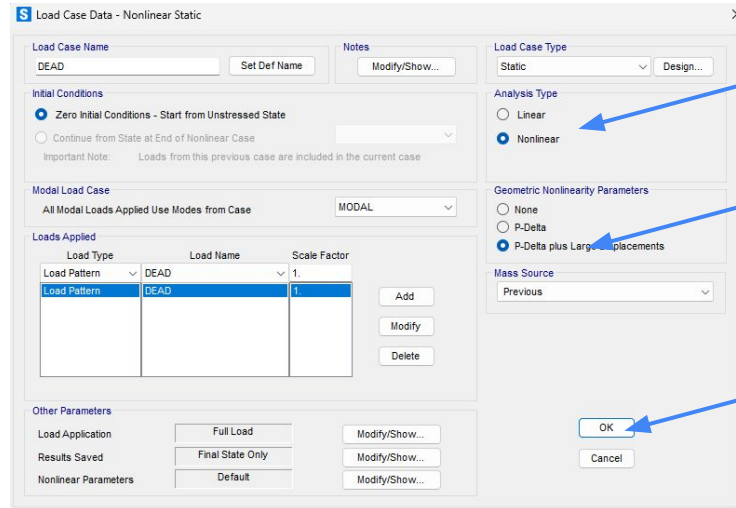


2.



Click Modify

3.



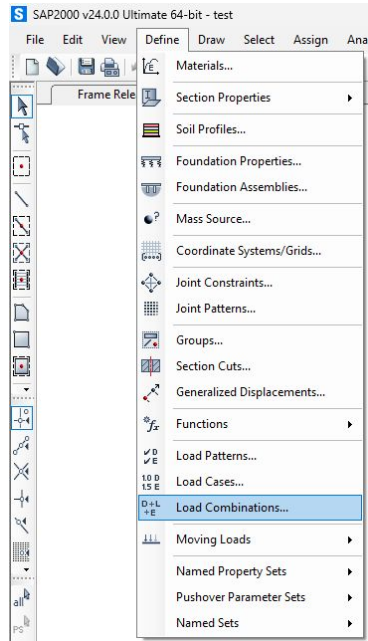
Non-linear

P-Delta + P-DELTA

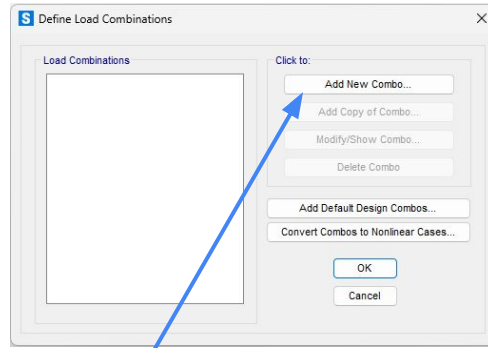
OK

# Load combinations

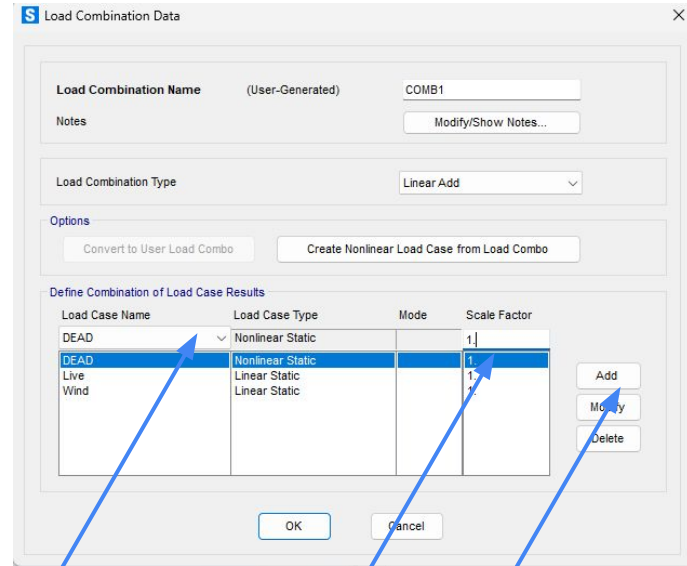
1.



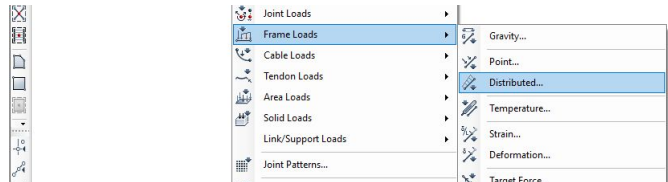
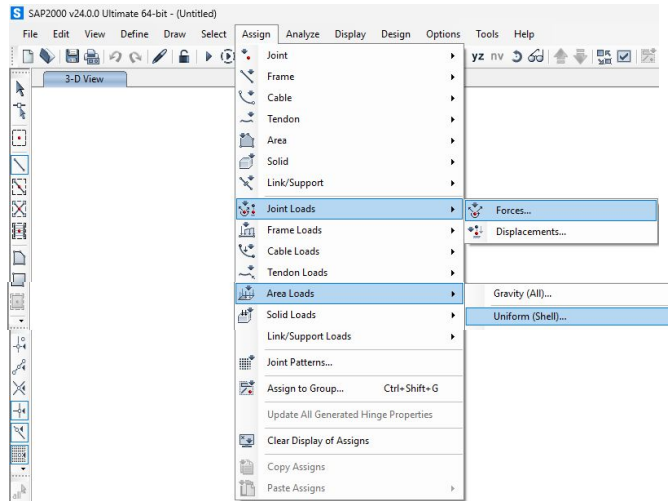
2.



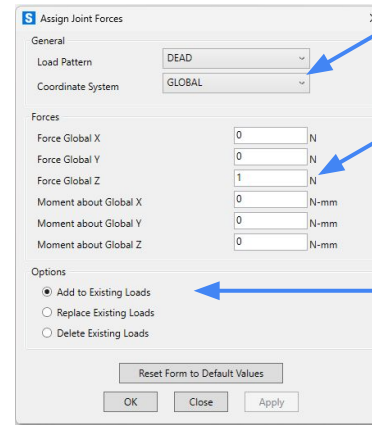
3.



# Assign loads



## Point Load



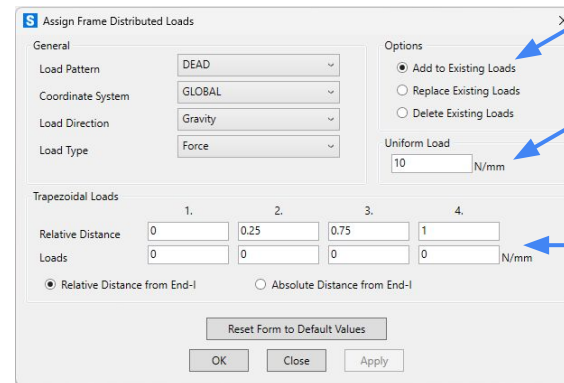
Choose load type

Add Force (Usually Z)

Make sure this is Add

=> Click on all points where you want the load  
=> OK

## Distributed Load



Make sure this is Add

Add distributed load

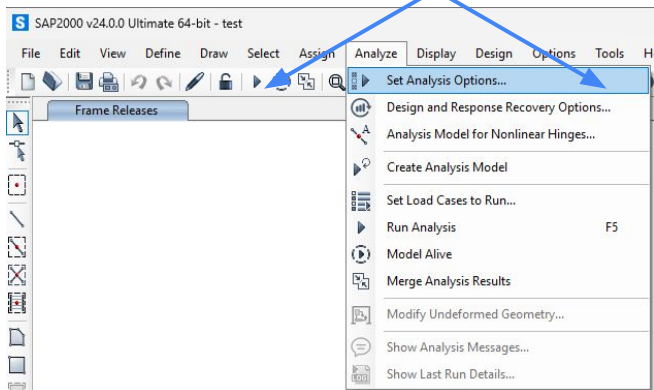
Only for non-uniform (Don't need to touch)

=> Click on all elements where you want the load  
=> OK

# Analysis

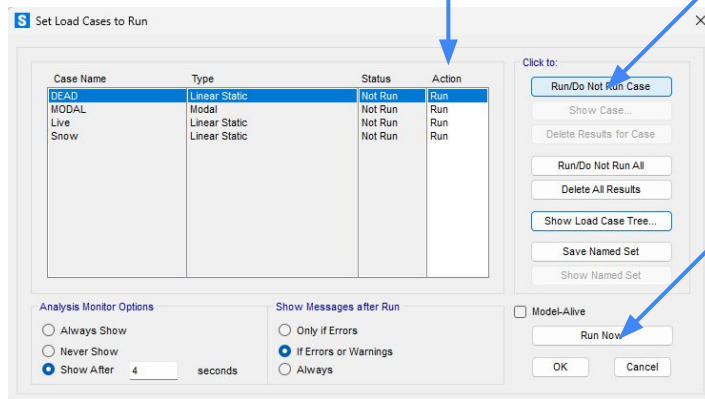
1.

Either



The ones that say  
“Run” will run

Click Run for all cases you  
want to see the results of

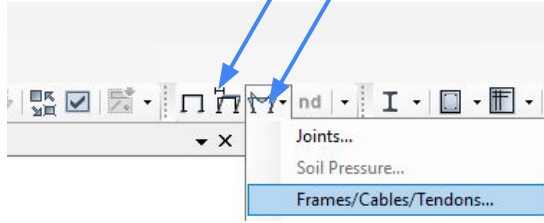


Click Run Now

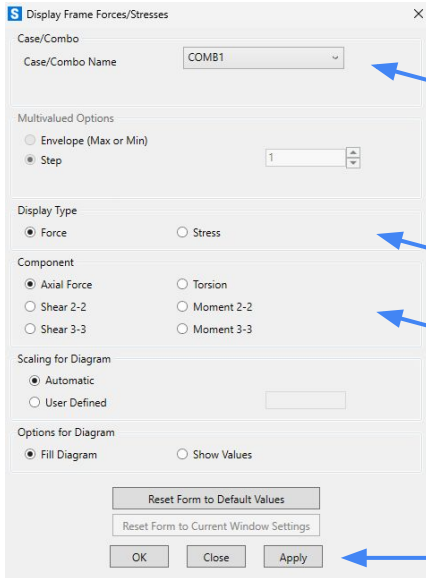
# View

Show Deformed shape

Show Internal Forces



Forces at Joint or Frame



Choose load/loadcombo

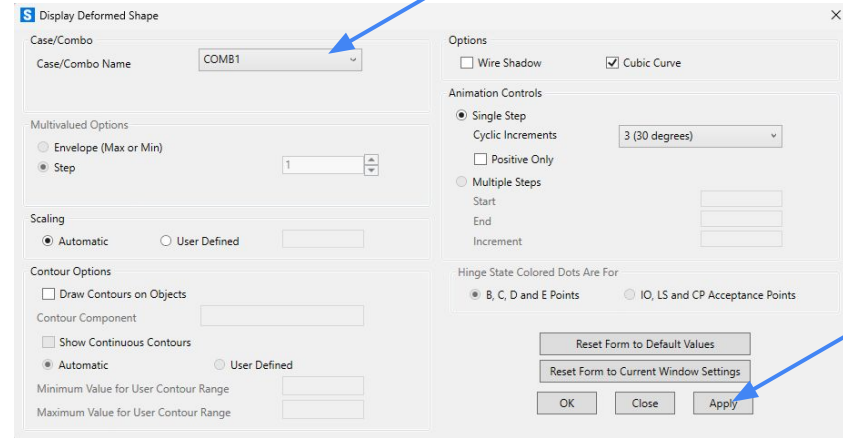
Force/Stress

Choose Force/Moment type

Apply

## Deformed shape

Choose load/loadcombo



Apply

# Dynamic Analysis

---

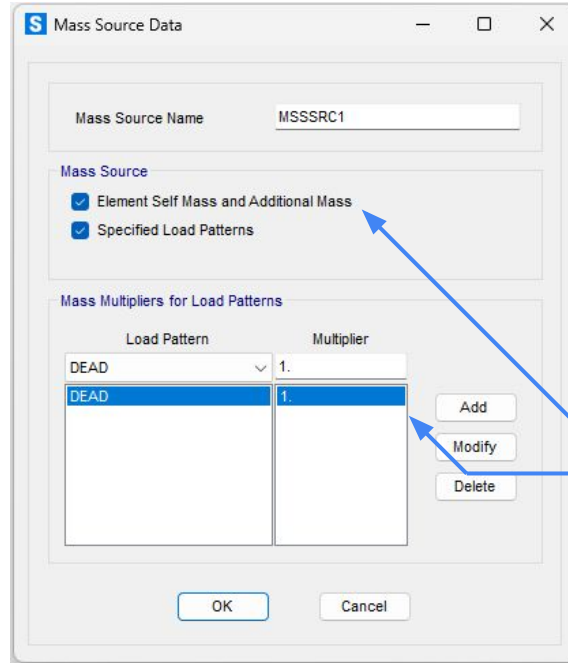
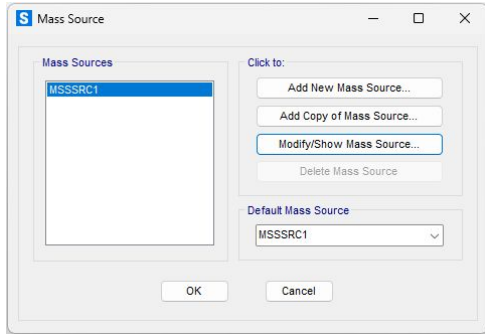
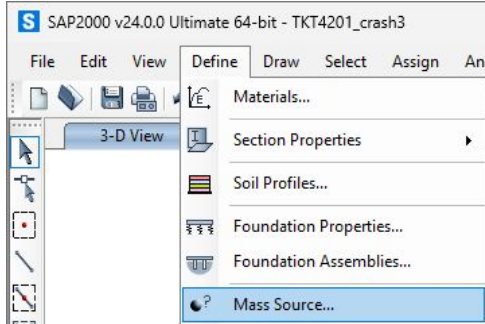
## Workflow RHA

1. Define Mass
2. Time History Function
3. Load Case
4. Deformed Shape

Video on Response Time History (RHA) analysis

[https://www.youtube.com/watch?v=VjkqA4\\_hlMk](https://www.youtube.com/watch?v=VjkqA4_hlMk)

# Define Mass Source

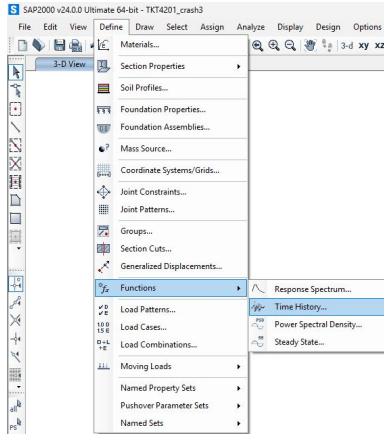


Default is usually correct

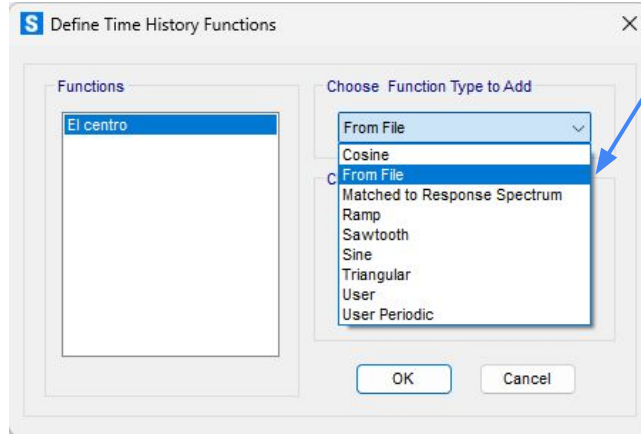
- **Element self Mass** includes mass of frame elements
- If you have additional masses add the load cases
  - Make sure the load cases don't have self weight =1 if Element self mass is turned on

Make sure self weight isn't duplicated

# Time History

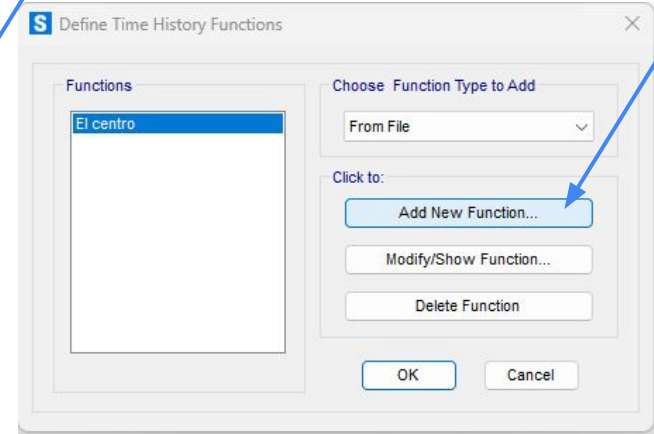


1.  
Define  
=> Functions  
=> Time history



2.

From File



3.



# Time History .TXT

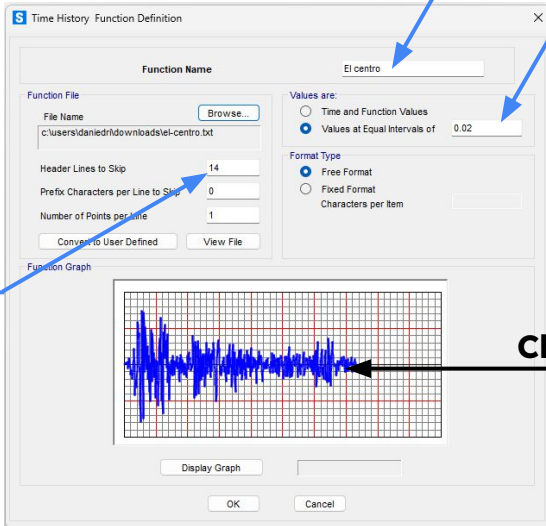
## TXT File 1

```
# Acceleration data for the north-south  
# El Centro, California, during the  
# The data is given in units of grav.  
# of 50Hz.  
# The time vector and data can be es  
# commands:  
#  
#   ug = numpy.loadtxt("el-centro.txt",  
#   dt = 1/50.  
#   t = np.arange(0, ug.size) * dt  
#  
# where 'ug' will be the ground accel.  
# in units of seconds.  
#  
0.00000  
0.00630  
0.00364  
0.00099  
0.00478
```

- Skip 14 rows
- $dt = 1/50 = 0.02$
- Acceleration values at equal intervals

Name

dt=0.02



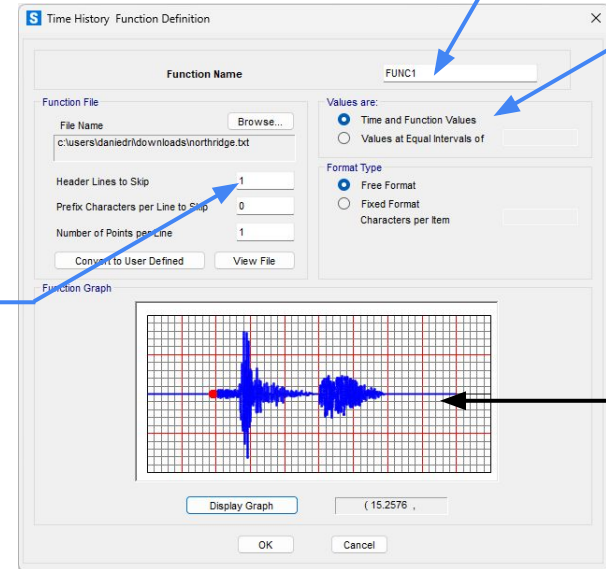
## TXT File 2

```
14383 0.005  
0.005 -2.8764145667603E-08  
0.01 -2.88162238514715E-08  
0.015 -2.88682488896783E-08  
0.02 -2.8920220030969E-08  
0.025 -2.89721365212768E-08  
0.03 -2.90239976037182E-08  
0.035 -2.90758025185872E-08  
0.04 -2.91275505033508E-08  
0.045 -2.91792407926437E-08  
0.05 -2.92308726182633E-08  
0.055 -2.92824452091646E-08
```

- Skip 1 row
- $dt = 0.005$
- Time & Acceleration values
- 14383 data points

Name

Time + a



# Define Earthquake Loads

**Name:** **Quake** **0** **Last: Click add**

The screenshot shows the 'Define Load Patterns' dialog box. It has a table with columns: Load Pattern Name, Type, Self Weight Multiplier, and Auto Lateral Load Pattern. The 'EQX' row is selected. A blue arrow points from the text 'Name:' to the 'EQX' cell. Another blue arrow points from 'Quake' to the 'Type' dropdown menu, which is open and showing 'Quake' as the selected option. A third blue arrow points from '0' to the 'Self Weight Multiplier' cell. A fourth blue arrow points from 'Last: Click add' to the 'Add New Load Pattern' button in the 'Click To:' section on the right.

| Load Pattern Name | Type      | Self Weight Multiplier | Auto Lateral Load Pattern |
|-------------------|-----------|------------------------|---------------------------|
| EQX               | Quake     | 0                      | None                      |
| DEAD              | Dead      | 0                      |                           |
| EQX               | Live      | 0                      | None                      |
| EQY               | Roof Live | 0                      | None                      |
|                   | Quake     |                        |                           |
|                   | Wind      |                        |                           |
|                   | Snow      |                        |                           |
|                   | SeaState  |                        |                           |
|                   | Other     |                        |                           |
|                   | More...   |                        |                           |

Click To:

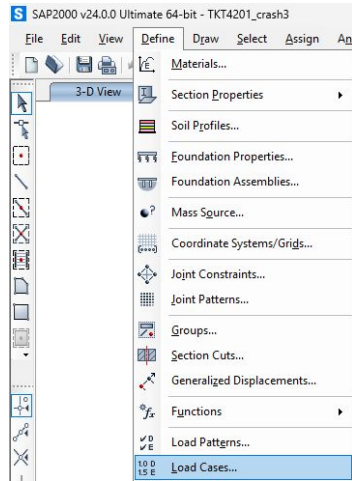
- Add New Load Pattern
- Add Copy of Load Pattern
- Modify Load Pattern
- Modify Lateral Load Pattern...
- Delete Load Pattern
- Show Load Pattern Notes...

OK Cancel

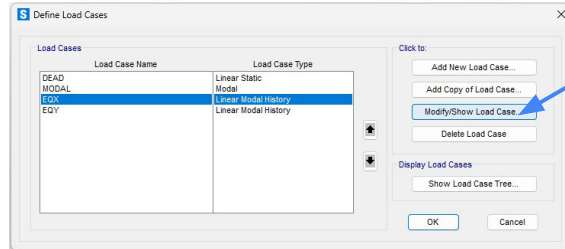
Add one load for each  
direction you want to consider

# Define Earthquake Loads

1.



2.

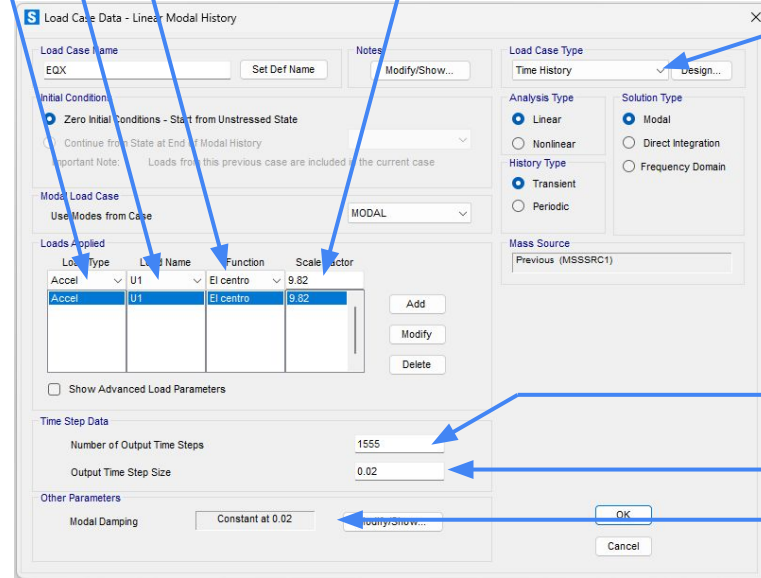


Function GM

U1=UX, U2=UY

Accel

3.



9.81

$1g = 9.81m/s^2$

Check units in TXT and bottom right, if units in TXT are in  $m/s^2$  scale factor = 1. If units of sap is in mm, scale factor = 9810.

Time History

Number of values in txt file

"dt" from txt file

$\zeta = 0.05$

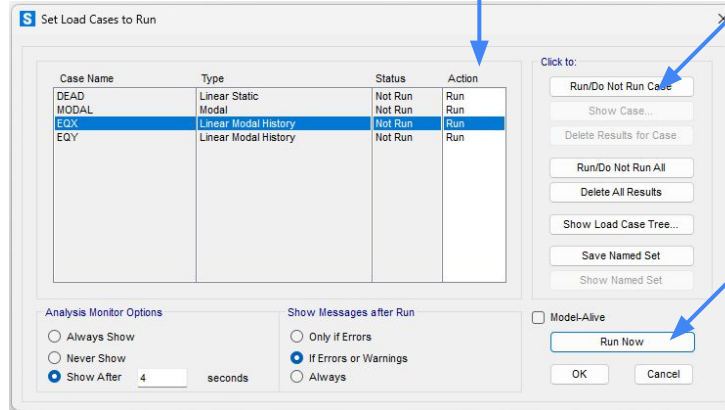
Other damping values should be used based on structural system

# Run analysis

1.



2.

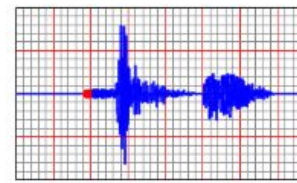


The ones that say  
"Run" will run

Click Run for all cases you  
want to see the results of

Click Run Now

# Show deformed shape



16sec → 55sec

Choose load  
Abs. Displace.

Time

Automatic

if you cant see the  
movement put user  
defined as 10, 50, 100

**S** Display Deformed Shape

Case/Combo  
Case/Combo Name: EQX  
Time History Display Type: Absolute Displacement

Multivalued Options  
☐ Envelope (Max or Min)  
☒ Time: 0 sec

Scaling  
☒ Automatic ☐ User Defined

Contour Options  
☐ Draw Contours on Objects  
Contour Component:   
☐ Show Continuous Contours  
☒ Automatic ☐ User Defined  
Minimum Value for User Contour Range:   
Maximum Value for User Contour Range:

Options  
☐ Wire Shadow ☒ Cubic Curve

Animation Controls  
☐ Single Step  
Cyclic Increments:   
☐ Positive Only  
☒ Multiple Steps  
Start Time: 0 sec  
End Time: 30 sec  
Time Increment: 0.1 sec

Hinge State Colored Dots Are For  
☒ B, C, D and E Points ☐ IO, LS and CP Acceptance Points

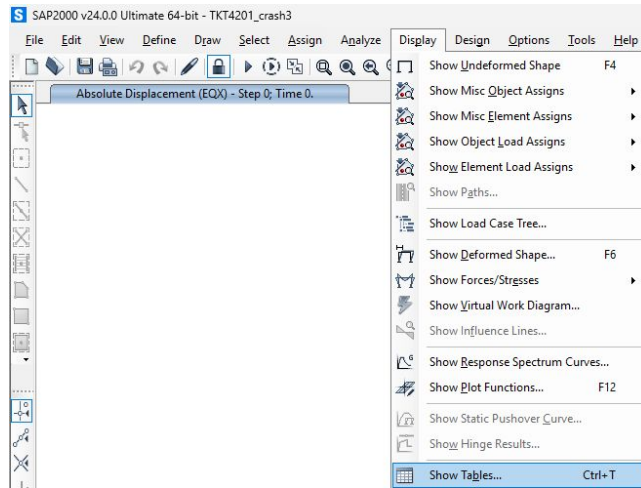
Reset Form to Default Values  
Reset Form to Current Window Settings  
OK Close Apply

Use values from  
txt

Last: Click Start  
Animation at  
bottom right

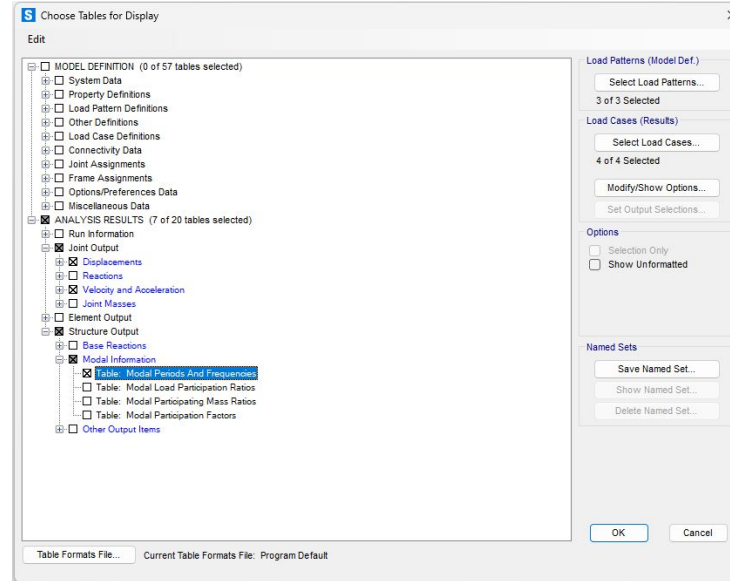
Optional: Adds  
colour to frame

# Read point values



1.  
Display  
=> Show Tables...

## Choose response quantities



Analysis Results  
→ Joint Output  
→ Displacements  
→ Velocity and Acceleration  
→ Structure Output  
→ Modal Periods and Frequencies

# Read point values

Click on header to see max/min

Joint index and load case

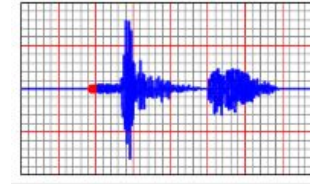
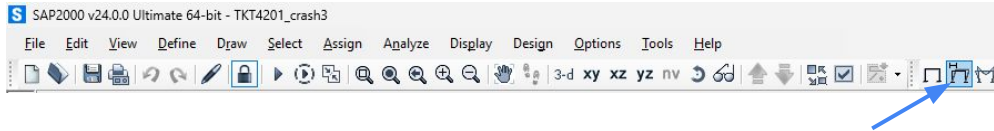
Choose Response

| Joint Text | Output Case | Case Type Text | Step Type Text | Step Num Unitless | U1 m      | U2 m      | U3 m      | R1 Radians | R2 Radians | R3 Radians |
|------------|-------------|----------------|----------------|-------------------|-----------|-----------|-----------|------------|------------|------------|
| 55         | EQX         | LinModHist     | Min            |                   | -0.385674 | -0.348243 | -0.004735 | -0.000682  | -0.006344  | -0.078645  |
| 77         | EQX         | LinModHist     | Min            |                   | -0.385674 | -0.272558 | -0.004153 | -0.00072   | -0.006332  | -0.078645  |
| 54         | EQX         | LinModHist     | Min            |                   | -0.361305 | -0.32749  | -0.004676 | -0.001944  | -0.011029  | -0.074528  |
| 76         | EQX         | LinModHist     | Min            |                   | -0.361303 | -0.253288 | -0.004088 | -0.001916  | -0.011066  | -0.074528  |
| 75         | EQX         | LinModHist     | Min            |                   | -0.330021 | -0.220124 | -0.003899 | -0.003236  | -0.015472  | -0.067583  |
| 53         | EQX         | LinModHist     | Min            |                   | -0.330015 | -0.286669 | -0.004503 | -0.002947  | -0.015657  | -0.067581  |
| 52         | EQX         | LinModHist     | Min            |                   | -0.305919 | -0.2323   | -0.004161 | -0.003485  | -0.017374  | -0.059415  |
| 74         | EQX         | LinModHist     | Min            |                   | -0.305898 | -0.186581 | -0.003527 | -0.004074  | -0.018333  | -0.059354  |
| 51         | EQX         | LinModHist     | Min            |                   | -0.291151 | -0.173059 | -0.003622 | -0.003735  | -0.015303  | -0.054058  |
| 73         | EQX         | LinModHist     | Min            |                   | -0.291102 | -0.167903 | -0.002943 | -0.003738  | -0.010871  | -0.040062  |
| 95         | EQX         | LinModHist     | Min            |                   | -0.291016 | -0.24451  | -0.001665 | -0.000689  | -0.008081  | -0.025452  |
| 72         | EQX         | LinModHist     | Min            |                   | -0.27775  | -0.162054 | -0.002363 | -0.00267   | -0.008283  | -0.036817  |
| 94         | EQX         | LinModHist     | Min            |                   | -0.27775  | -0.223066 | -0.001522 | -0.00197   | -0.011173  | -0.020183  |
| 50         | EQX         | LinModHist     | Min            |                   | -0.277668 | -0.114519 | -0.002948 | -0.003345  | -0.007917  | -0.031243  |
| 28         | EQX         | LinModHist     | Min            |                   | -0.277532 | -0.120044 | -0.000361 | -0.001069  | -0.007409  | -0.015206  |
| 66         | EQX         | LinModHist     | Min            |                   | -0.266426 | -0.348179 | -0.001458 | -0.000682  | -0.001809  | -0.078633  |

Record: 1 of 1496

Add Tables... Done

# Show Modal shapes



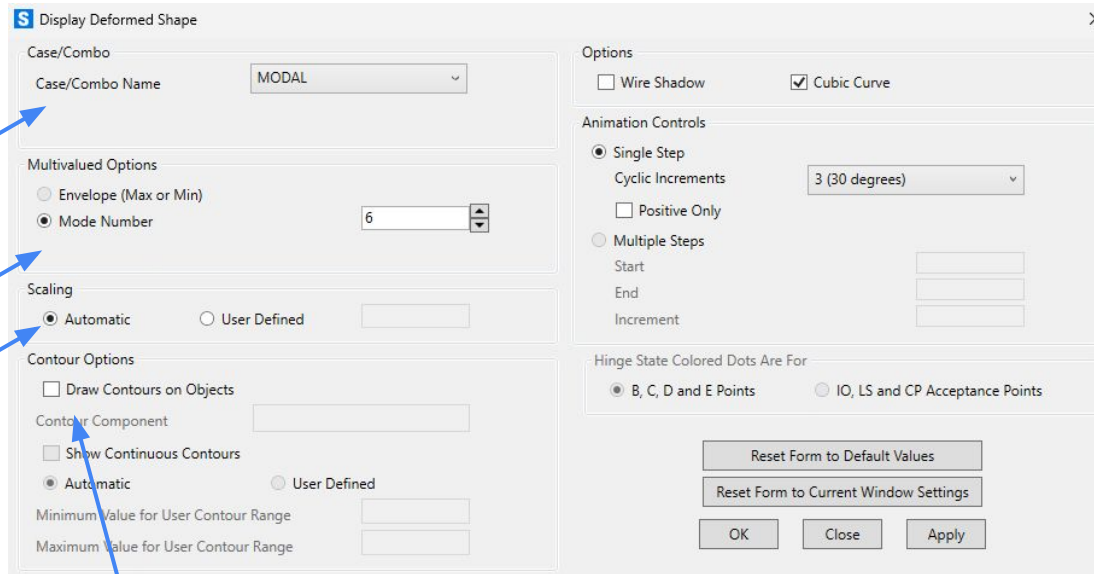
16sec → 55sec

Choose load

Choose Mode #

Automatic

if you cant see the movement put user defined as 10, 50, 100



Optional: Adds colour to frame

Last: Click Start Animation at bottom right